

NOS 110 – Operating Systems Concepts Syllabus

Course Information

Computer Technologies Division
Networking and Computer Technologies

Course Description:

This course introduces students to a broad range of operating system concepts, including installation and maintenance. Emphasis is placed on operating system concepts, management, maintenance and resources required. Upon completion of this course, students will understand OS concepts, installation, management and maintenance, using a variety of operating systems.

Credit Hours: Three (3) Semester Hours

Prerequisite: none

Student Learning Outcomes

Upon completion of this course, students will be able to:

1. Students will demonstrate knowledge of general operating system concepts.
2. Students will install operating systems using virtualization software.
3. Students will demonstrate a mastery of navigation, folder/directory structure creation, security, and file management in both a GUI and command line environment.
4. Students will demonstrate the ability to run basic Windows and Linux Command Line Interface commands using the correct switches and parameters to obtain specific information and results.
5. Students will demonstrate the ability to create and execute basic Windows and Linux Command Line Interface script files.
6. Students will set operating system preferences.
7. Students will observe the operating system for basic network interactions by sharing files, disks, and printers.

Course Requirements and/or Expectations

Software Requirement: Word processing (Microsoft Office, Libre Writer, or Open Office, VMware Workstation or Fusion (provided by Wake Tech.)

Hardware Requirement: Storage device for file transfer (labs, iso files, virtual machines). USB drive 12-GB (16-GB recommended) or higher, external hard drive, or cloud storage

Special Instructions:

This course requires students to use VMware virtualization software. The software is provided to the student free of charge. Windows, Mac, and Linux versions of VMware are available. Students are expected to do work outside of the classroom hours if taking a seated/hybrid course.

Therefore, **students must have a Windows computer that supports VMware, or be able to access a Windows computer that supports virtualization.** A minimum of **12GB of RAM memory is required, 16GB of RAM memory is recommended.** Campus computers are available, and it is the student's responsibility to use those resources. **Online students must adjust personal computers to accomplish the coursework.**

Students will also receive a Wake Tech *Microsoft Azure for Education* account for downloading Windows iso files. This account is provided to the student free of charge.

Required Textbooks

Jane Holcomb and Charles Holcombe

Survey of Operating Systems, 7th edition

McGraw-Hill, 2023, e-book (Free with Eagle Advantage, ISBN – 978-1-265128913)

McGraw-Hill, 2023, Loose-leaf Option (\$30, ISBN – 978-1-26569174)

Grading or Evaluation

Labs and Assignments:	30%
Quizzes:	15%
Midterm Exam:	15%
Final Exam:	15%
Windows Skills Exam:	10%
Linux Skills Exam:	10%
Smart Book Assignments:	3%
Employability Skills:	2%

Testing Requirement:

Online students must take the Quizzes, Midterm and Final Exams using **Respondus** proctoring software, which may require a webcam. Instructions are provided in the course shell on downloading, installing, and using **Respondus**.

All Privacy information for **Respondus** can be found here: <https://web.respondus.com/privacy-policy/>.

Respondus software strictly adheres to all data privacy & security laws & standards.

Disability Support Services

Wake Technical Community College does not discriminate based on disability in the admissions or employment processes or in access to programs, facilities, or activities. Disability Support Services (DSS) is available for students who require academic accommodations due to any physical, psychological, or learning disability. Please visit the website for Disability Support Services (opens in new window), URL: [<http://disabilityservices.waketech.edu>], for additional information. To connect with a DSS coordinator and determine eligibility, contact the Main campus at Phone: 919-866-5670 (Sorenson video phone for the deaf: 919-324-1508). Fax: 919-662-3616. Or, contact North campus office at Phone: 919-532-5713. Fax: 919-878-1896.

Wake Tech Core Values

Wake Technical Community College will structure its operations, training and educational programs around the Core Values of accountability, respect, responsibility, critical thinking, communication, and collaboration.

- **Accountability**—Accountability is essential for an environment of learning. Those who are accountable stand by their words and actions, taking full responsibility for what they create and for what they contribute to the community.
- **Respect**—Respect is a prerequisite for enhancing learning. Community members who respect themselves and others help create a safe, yet open, climate of learning.
- **Responsibility**—Responsibility is the root of success. Students who assume personal responsibility for their education will reach their goals. Responsible students also make contributions to their communities.
- **Critical Thinking**—Critical thinking is the fundamental purpose of higher education. The ability to solve problems through the application of the appropriate skills is critical to all disciplines.
- **Communication**—Communication is increasingly the key competency for living and working in the information age. Communicating effectively in oral and written forms through traditional and new media is a powerful tool for personal and career success.
- **Collaboration**—Collaboration, by bringing together individual knowledge and talents, creates teams that are greater than the sum of their parts. Such teamwork maximizes benefits to individuals and the community.

Policies

Students are responsible for reading the information and policies set forth in the [Wake Technical Community College catalog \(opens in new window\)](http://catalog.waketech.edu), URL: [http://catalog.waketech.edu], as well as additional requirements set by the instructor. Students are held accountable for this information. New policies and policy changes will be communicated to students on the official [Updates page \(opens in new window\)](http://Updates.waketech.edu), URL: [http://Updates.waketech.edu].

Communication

Appropriate Communication—Students are expected to communicate in clear, correct, respectful Standard American English at all times. This includes all online or electronic communication, which should follow the standards of Internet etiquette (also known as “netiquette”). Students unfamiliar with these concepts and conventions should check with their instructors.

Attendance Policy

Absences—The college catalog student handbook states that “students are expected to be in attendance at least 90 percent of all scheduled class meetings. An absence is defined as missing one-third or more of any regularly scheduled class meeting.” If you anticipate an absence, notify your instructor in advance. “If prior notification is not possible, the student should contact the instructor immediately upon returning to the College to determine the next course of action.”

Absences due to religious holidays—The college catalog states that “Wake Tech students are allowed two class days of excused absences per academic year for religious observances. It is the student's responsibility to contact the instructor for each course in which work will be missed. The student must provide written notification to the instructor within the first two weeks of the semester, identifying the religious observance and date of the planned absence.”

Days missed for religious observances will be counted as absences, but they will not be considered part of the allowed 10%. Reasonable accommodation will be made for affected students in regard to any resulting missed work.

Verification of Attendance in Online or Hybrid Courses—In order to remain enrolled in an online or hybrid course, each student is required to open and complete the Course Entry Quiz in Blackboard by 11:59 PM on the semester 10% date. Students should complete the activity as soon as possible. Delay in completion may result in getting behind in coursework or in missing assignments.

Students who do not complete the Course Entry Quiz by the deadline will be dropped from the course with no tuition refund.

Late Work Policy

Each project/assignment/exercise will come with specific instructions for completion. All assignments/class work **MUST** be completed (if applicable)! It is your responsibility throughout the semester to keep up with your course work. It is not the responsibility of the instructor to advise you of missed work during the term or during the last week of the term. Employability Skills are directly correlated to assignment and assessment completion and submission prior to or by the posted due date.

Refer to the **Course Resources** section within your Blackboard course shell for the late work policy established by your course instructor.

Exams must be taken on the assigned date; otherwise, a grade of zero will be assigned. Any exception to this policy must be the result of a documented emergency or prior arrangement with the instructor. There will be no exceptions to this rule.

ANY MAKEUP EXAMS must be taken prior to the scheduled exam date. This must be pre-arranged with the instructor. Last minute arrangements are not acceptable.

Official Communication with Students Policy

Every curriculum student is provided with an official Wake Tech email account through the [Wake Tech student portal \(opens in new window\)](http://my.waketech.edu), URL: [http://my.waketech.edu].

- Students must first activate their my.waketech.edu account, wait 24 hours, and then activate the email account.
- This college-issued email account is to be used for all email correspondence with instructors and other college officials.

Official correspondence from the college (communications from instructors, information about registration or financial aid, etc.) will be sent to students' Wake Tech email address **ONLY**. Instructors and college officials may refuse to accept student emails sent from other addresses. Review the Registration and Records section in [Wake Tech's catalog \(opens in new window\)](http://catalog.waketech.edu), URL: [http://catalog.waketech.edu], for updates to the Official Communication with Students Policy.

FERPA Statement

Review the Academic Policies section of [Wake Tech's catalog \(opens in new window\)](http://catalog.waketech.edu), URL: [http://catalog.waketech.edu], for updates.

Following the *Family Educational Rights and Privacy Act (FERPA) of 1974*, instructors are prohibited from discussing a student's grade or other matters related to academic performance outside of the Blackboard grade center. Faculty may send grades to individual students via email only when written authorization from the student is on file (WTCC Form 1126).

The Authorization to Release Instructor Graded Material (WTCC Form 1126) does not apply to final grades. Final grades will be accessed through Web Advisor only or with a Request for Transcript. Students may request this form from the instructor.

Wake Technical Community College Academic Integrity Policy

“Wake Technical Community College expects students to demonstrate the highest personal integrity in all academic work and behavior.... Students may be asked to sign a statement of academic integrity upon entering Wake Tech classes.” Review the Student Code of Conduct, Rights, and Responsibilities section of [Wake Tech’s catalog \(opens in new window\)](http://catalog.waketech.edu), URL: [http://catalog.waketech.edu], for additional information.

Student Code of Conduct

“Students are expected to conduct themselves in accordance with generally accepted standards of scholarship and conduct. The purpose of the Student Code of Conduct (the Student Code) is not to restrict student freedom but to protect the rights of all students in their academic pursuits.” Review the Student Code of Conduct, Rights, and Responsibilities section of [Wake Tech’s catalog \(opens in new window\)](http://catalog.waketech.edu), URL: [http://catalog.waketech.edu], for additional information.

Schedule (The following is a tentative schedule and subject to change)

Note: The order in which these subject areas are presented may be changed/modified by your instructor. This list is offered only as a guide. The pace of each class differs according to the instructional needs of the students in the class. Always consult with your instructor. **Be sure to follow the weekly assignment posts within Blackboard.**

Schedule (tentative)			
Week	Lesson	Topics and/or Assignments	Textbook
1	1	Course Entry Quiz Respondus Practice Quiz Introduce Yourself Discussion Board Chapter 1 reading Chapter 1 labs as assigned Chapter Quiz	Ch 1 - Introduction to Operating Systems <i>Establish Microsoft Azure for Education Account</i>
2	2	Chapter 11 reading Chapter 11 labs as assigned Number Conversions (Dec, Bin) Chapter Quiz	Ch 11-File Management in the Cloud <i>Number System Conversions</i>
3	3	Chapter 3 reading VMware Registration Chapter 3 labs as assigned Number Conversions (Dec, Bin, Hex) Chapter Quiz	Ch 3-Desktop Virtualization <i>Number System Conversions continued</i>
4	4	Chapter 4 reading Download Windows 10 iso. Install Windows 10 Chapter 4 labs as assigned Number Conversion Quiz	Ch 4-Windows 10 <i>Students must have a valid Azure for Education account at this point</i>
5	5	Chapter 4 reading <i>continued</i> Chapter 4 labs as assigned Chapter Quiz	Ch 4-Windows 10
6	6	Chapter 5 reading Chapter 5 labs as assigned Chapter Quiz	Ch 5- Supporting & Troubleshooting Windows
7	7	MIDTERM EXAM (Chapters 1, 3, 4, 5, Conversions)	15% of Final Grade
7-8		Chapter 9 4 th edition Reading Chapter 9 4 th edition labs as assigned Chapter Quiz	Ch 9-Holcombe 4th edition (pdf available) Windows Command Line Interface
9		Windows SKILLS Review & Exam Windows Skills Exam Prep Assignment <i>Optional</i> TLC Skills Exam Review Windows Skills Exam (Hands-on skills testing based on Windows CLI labs and exercises)	10% of Final Grade
10		SPRING BREAK	

Schedule (tentative) continued			
Week	Lesson	Topics and/or Assignments	Textbook
11	8	Chapter 12 Reading Chapter 12 labs as assigned Win 11 Key Terms Discussion Board Chapter Quiz	Ch 12-Windows 11 (eBook only)
12	9	Chapter 7 Reading Chapter 7 labs as assigned	Ch 7-Linux on the Desktop <i>Linux Basic Commands handout (required reading)</i>
13	10	Chapter 7 Reading <i>continued</i> Chapter 7 labs as assigned Chapter 7 Quiz	Ch 7-Linux on the Desktop <i>Introduction to VI handout (required reading)</i>
14	11	Linux Command Line Interface Linux CLI labs as assigned Linux-Windows CLI Discussion Board	Linux Command Line Interface Not in textbook <i>Supplemental Reading Material (required reading)</i>
15		Linux SKILLS Review & Exam Linux Skills Exam Prep Assignment <i>Optional</i> TLC Skills Exam Review Linux Skills Exam (Hands-on skills testing based on Linux CLI labs and exercises)	10% of Final Grade
16	12	Chapter 9 Reading Chapter 9 labs as assigned Chapter Quiz	Ch 9-Connecting Desktops and Laptops to Networks
		FINAL EXAM Chapters 9 (4 th ed), 7, 9, 12	15% of Final Grade